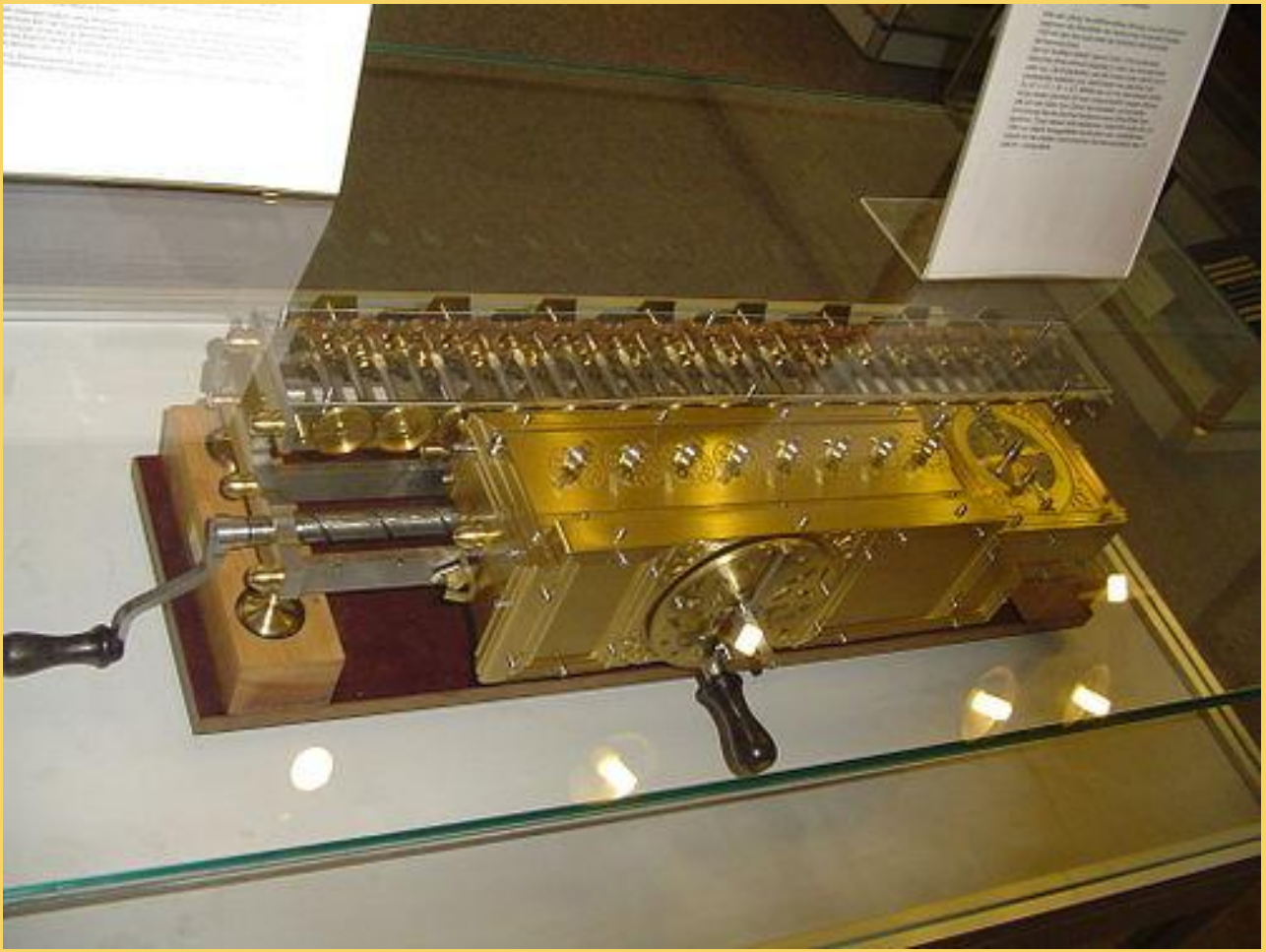


HW 6 due next Monday, Mar 23.

Quiz 8 due next Monday.

Class 17: Turing Completeness

University of Virginia
CS3120: DMT2
https://weikailin.github.io/cs3120-toc
Wei-Kai Lin

Plan

Recap Midterm 2: Circuits

Turing Machines

Random-Access Machines

Turing Completeness

Computability

Textbook [TCS] Section 7 and 8

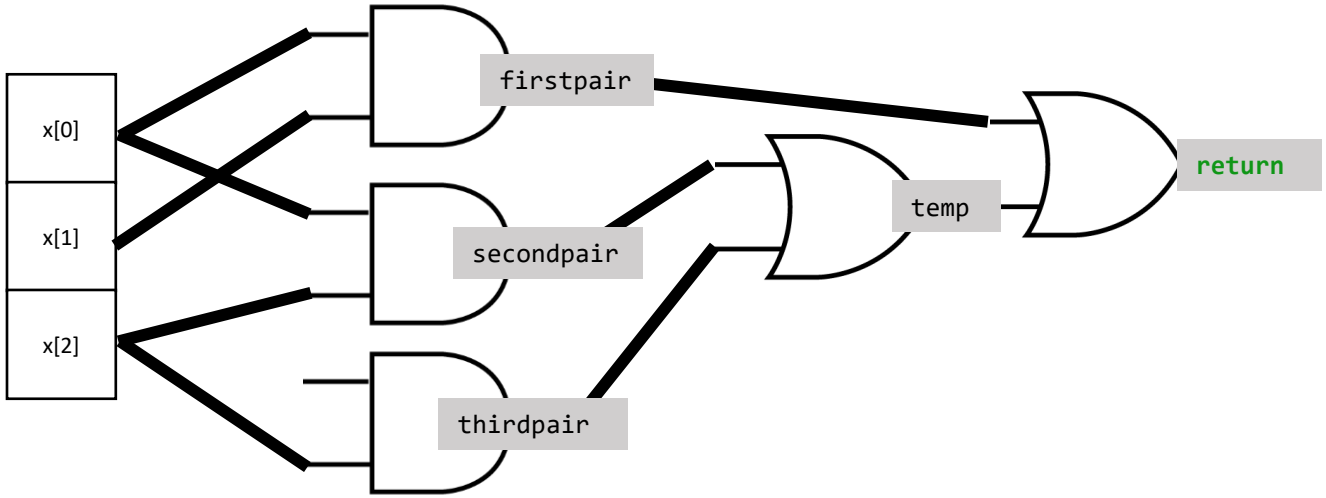
https://introtcs.org/public/lec_06_loops.html
Loops and infinity

Circuit: Built from “Boolean Gates”

A set of gates: {AND, OR, NOT}

Another set: {NAND}

Written as a DAG, or a “straightline program”




```
def MED(X[0],X[1],X[2]):
    firstpair = AND(X[0],X[1])
    secondpair = AND(X[1],X[2])
    thirdpair = AND(X[0],X[2])
    temp = OR(secondpair,thirdpair)
    return OR(firstpair,temp)
```

Universal Set of Gates

{AND,OR,NOT} is universal.

{NAND} is universal.

{AND,OR} is Not.

Monotone

Theorem 3.12 (NAND is a universal operation)

For every Boolean circuit C of s gates, there exists a NAND circuit C' of at most $3s$ gates that computes the same function as C.

Definition 3.20 (General straight-line programs)

Let $\mathcal{F} = \{f_0, \ldots, f_{t-1}\}$ be a finite collection of Boolean functions, such that $f_i : \{0,1\}^{k_i} \to \{0,1\}$ for some $k_i \in \mathbb{N}$. An $\mathcal{F}$ *program* is a sequence of lines, each of which assigns to some variable the result of applying some $f_i \in \mathcal{F}$ to k_i other variables. As above, we use `X[` i `]` and `Y[` j `]` to denote the input and output variables.

We say that $\mathcal{F}$ is a *universal set of operations* (also known as a universal gate set) if there exists a $\mathcal{F}$ program to compute the function $NAND$.

Circuit Size, and Class SIZE(s)

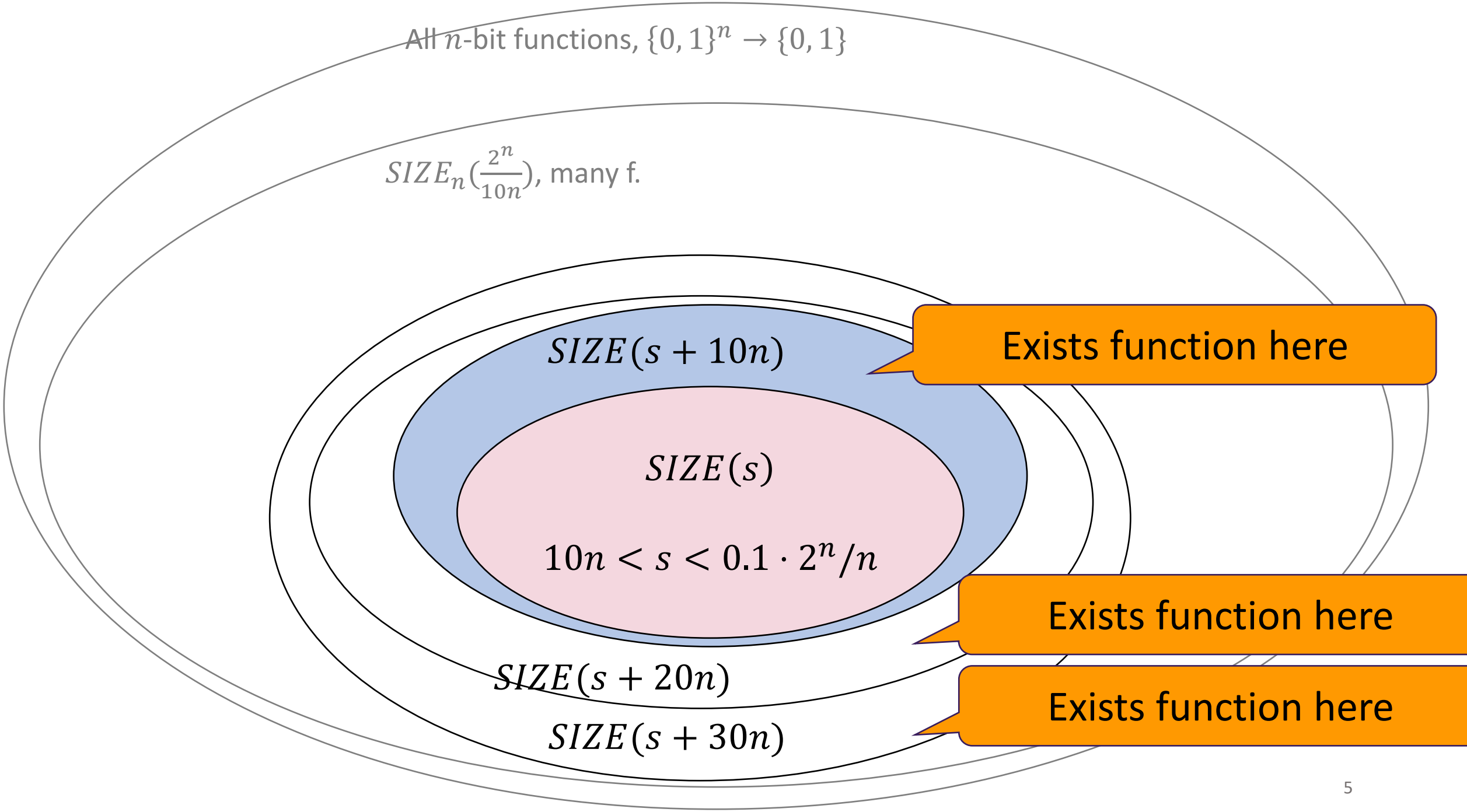
Circuit size: number of gates in a circuit

SIZE(s): the set { function f | there exists s-gate circuit such that computes f}

Is circuit $C \in SIZE(s)$?

$\in EVEN$?

Category Error!

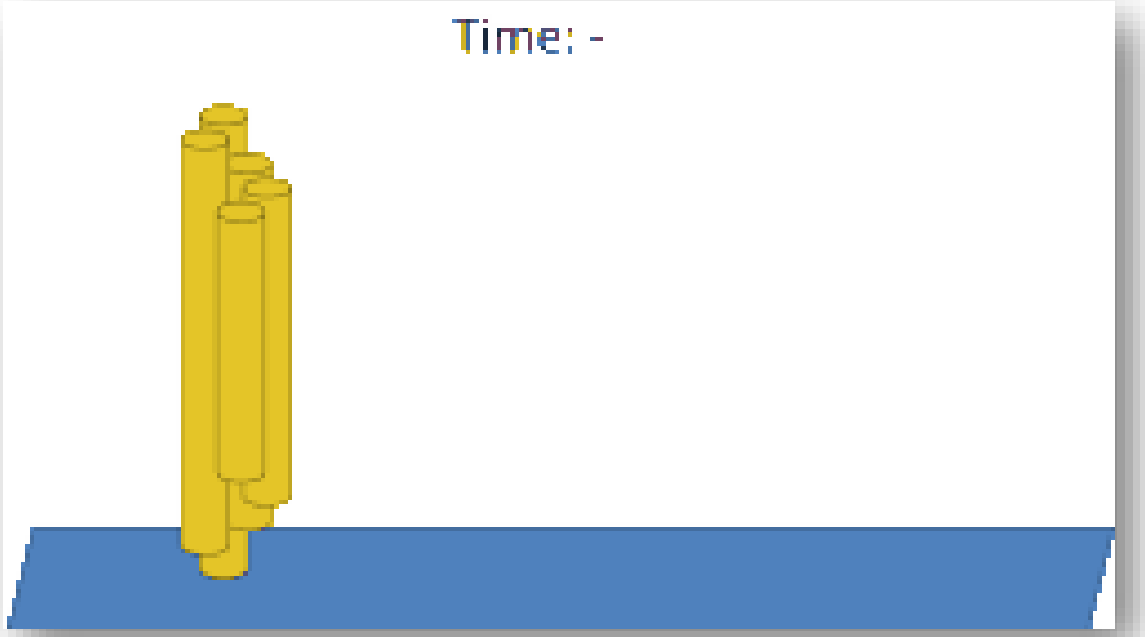

All n-bit functions, $\{0,1\}^n \to \{0,1\}$
$SIZE_n(\frac{2^n}{10n})$, many f.
$SIZE(s+10n)$
$SIZE(s)$
$10n < s < 0.1 \cdot 2^n/n$
$SIZE(s+20n)$
$SIZE(s+30n)$
Exists function here
Exists function here
Exists function here

Quiz 7: Spaghetti Sort

"Physical Extended Church-Turing Thesis" (PECTT): *If a function* $F : \{0,1\}^n \to \{0,1\}^m$ *can be computed in the physical world using* s *amount of "physical resources" then it can be computed by a Boolean circuit program of roughly* s *gates.*

Contradiction?

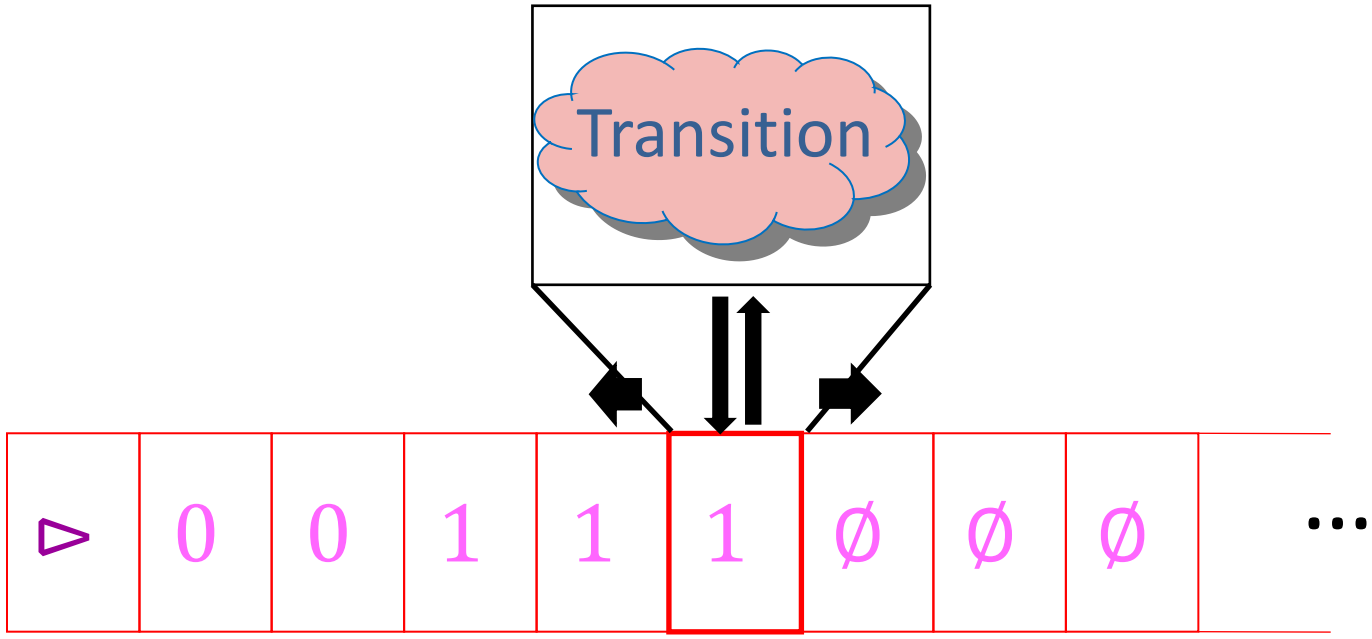
Why not?



https://en.wikipedia.org/wiki/Spaghetti_sort

Two-Tape TM and Random-Access Machines

Recap: Turing Machine



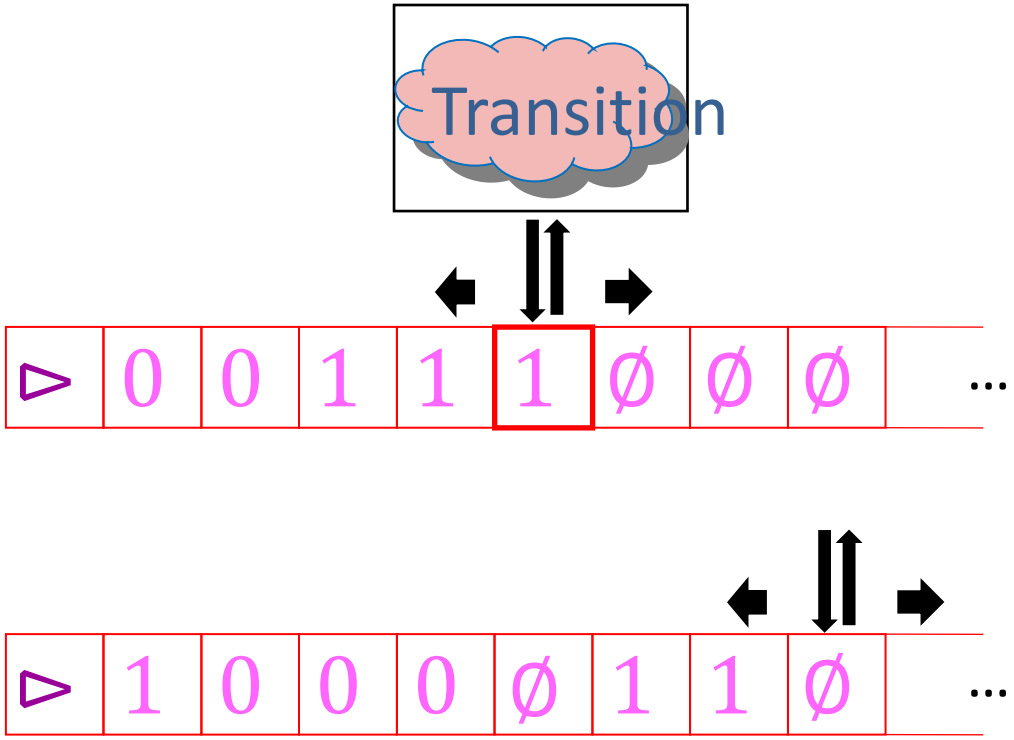
Tape initially contains the input followed by blanks

If transition "halts", → tape contains the output

Semi-infinite tape (memory)

Recap: Two-Tape Turing Machine

The transition reads, writes, and moves two tapes arbitrarily.



Recap: Random-Access Machine (RAM)?

A *Turing Machine,* is defined by (Σ, C, δ):

$C \in \mathbb{N}$: a finite number of states

Σ : alphabet – finite set of symbols

$$\Sigma \supseteq \{0, 1, \triangleright, \emptyset\}$$

$\boldsymbol{\delta}$: transition function

$$\boldsymbol{\delta}: [C] \times \Sigma \rightarrow [C] \times \Sigma \times \{\mathbf{L}, \mathbf{R}, \mathbf{S}, \mathbf{H}\}$$

A *RAM* is defined by (Σ, C, δ):

$C \in \mathbb{N}$: a finite number of states

Σ : alphabet – finite set of symbols

$$\Sigma \supseteq \{0, 1, \triangleright, \emptyset\}$$

$\boldsymbol{\delta}$: transition function

$$\boldsymbol{\delta}: [C] \times \Sigma \rightarrow [C] \times \Sigma \times \{\mathbf{L}, \mathbf{R}, \mathbf{S}, \mathbf{H}\} \times \mathbb{N}$$

Only difference.
What's wrong?

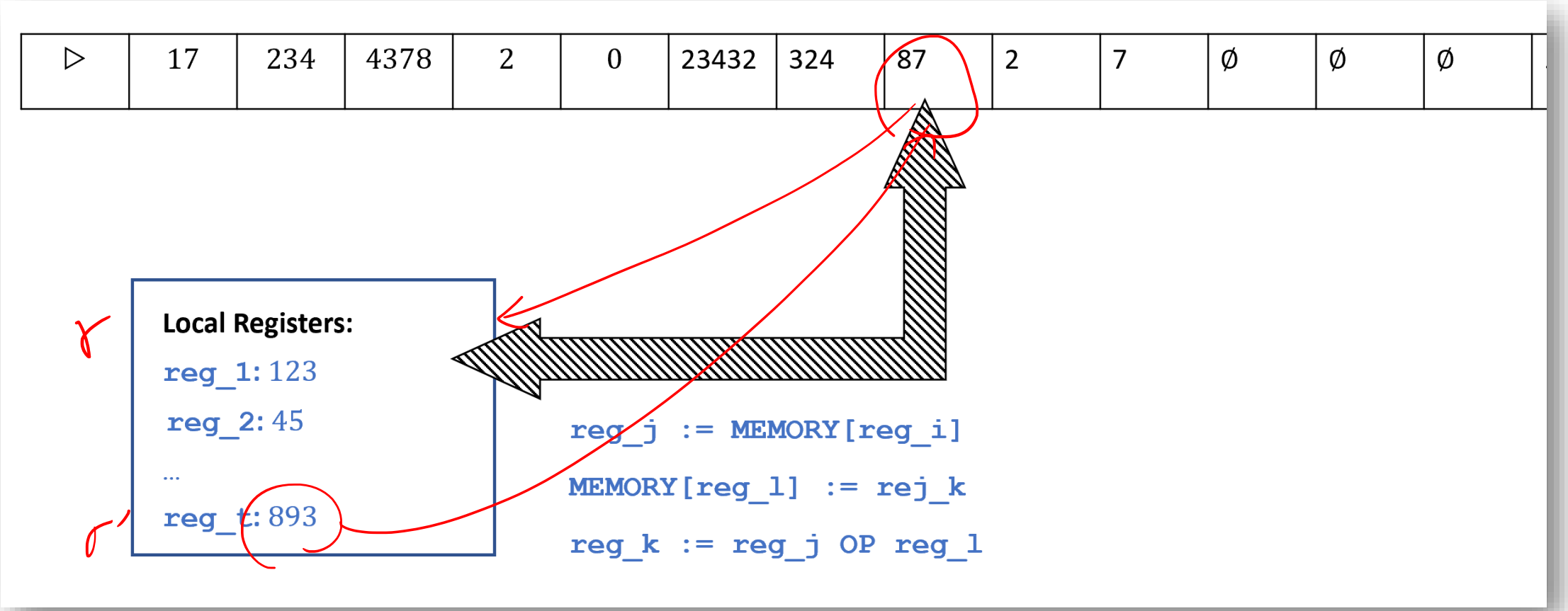


Figure 8.2: A *RAM Machine* contains a finite number of local registers, each of which holds an integer, and an unbounded memory array. It can perform arithmetic operations on its register as well as load to a register r the contents of the memory at the address indexed by the number in register r'

https://introtcs.org/public/lec_07_other_models.html

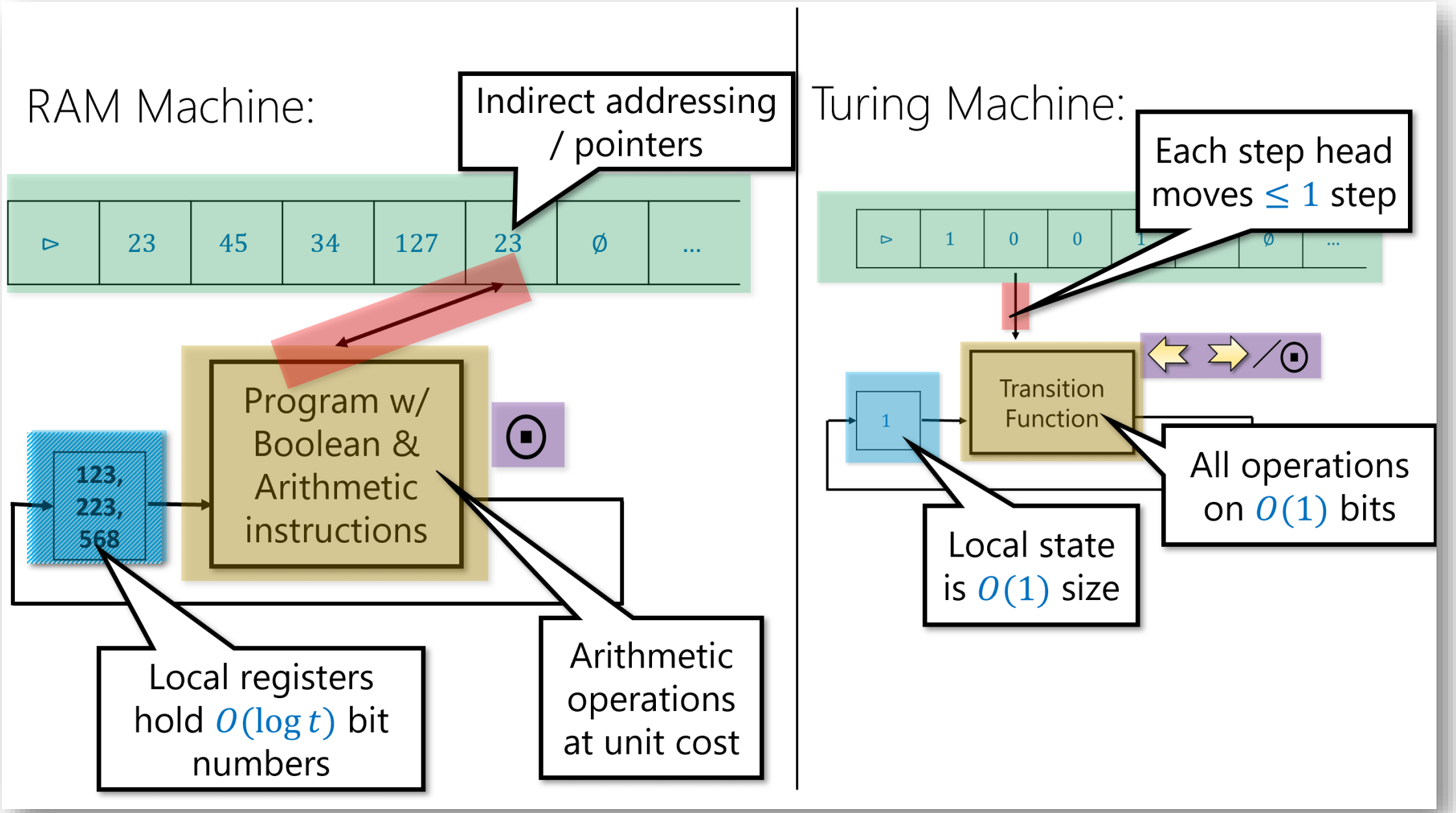


Figure 8.3. Also, "t is the number of steps that have been executed by the program so far"

WK's Def: Random-Access Machine (RAM)

A *RAM* is defined by a Two-Tape Turing Machine, but it is augmented with **J** (jump) in addition to L/R/S/H.
During execution on J, Tape2 is interpreted as a (potentially negative) integer k, and Tape1 is moved by k grids.

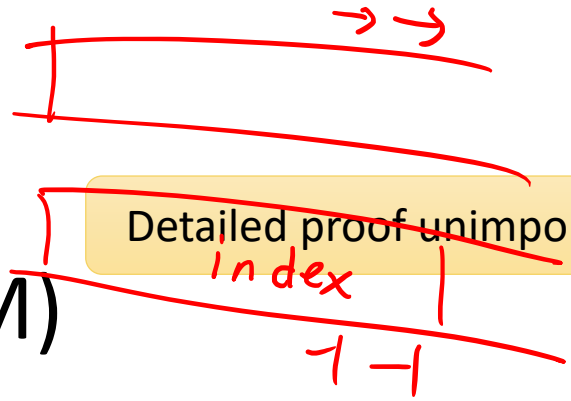
Another def: jump no more than B distance for some finite B.

Detailed definition unimportant

RAM is equivalent to One-Tape TM

Easy: RAM $\cancel{\subseteq}$ $\supseteq$ Two-Tape-TM

Doable: RAM $\subseteq$ Two-Tape-TM
(need to simulate any RAM by two-tape TM)



Detailed proof unimportant

RAM $\equiv$ Two-Tape-TM

$\equiv$ One-Tape-TM

Turing Completeness

Consequences of RAM ≡ TM

inf loop & array

easy : PL ⊇ TM

Any prog language PL that runs in the RAM model:
PL ⊆ TM.

PL includes:

Python, C/C++, Java, …, Haskel, OCaml, lambda-calculus, Conway's Game of Life, …,

DFA, NFA, Regular Expressions, Circuits.

Turing Complete

For any computation model X (or programming lang), we say X is *Turing Complete* if and only if X ≡ TM.

e.g. "Smart router" switch

Turing complete X includes:

Python, C/C++, Java, …, Haskel, OCaml, lambda-calculus, Conway's Game of Life, …,

(Ideal-)Python ≡ TM?

Also: C/C++, Java, …, Haskel, OCaml ….

What does it mean?

"ONLY" ~~capap~~ capability same.

resource differ

Write Python, compile to RAM to Two-tape to one tape

Computability

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]

The "computable" numbers may be described briefly as the real numbers whose expressions as a decimal are calculable by finite means. Although the subject of
it is almost equally easy
of an integral variable
predicates, and so fort
however, the same in eac
for explicit treatment as
shortly to give an account of the relations of the computable numbers, functions, and so forth to one another. This will include a development of the theory of functions of a real variable expressed in terms of computable numbers. According to my definition, a number is computable if its decimal can be written down by a machine.

Roughly: numbers that can be output *arbitrarily precisely* by a Turing Machine (No Input model).

Computable Numbers

Is $\frac{1}{2}$ *computable?*

0.5 0000 . . . .

for all i, output ith bit of 0.5000...
digit

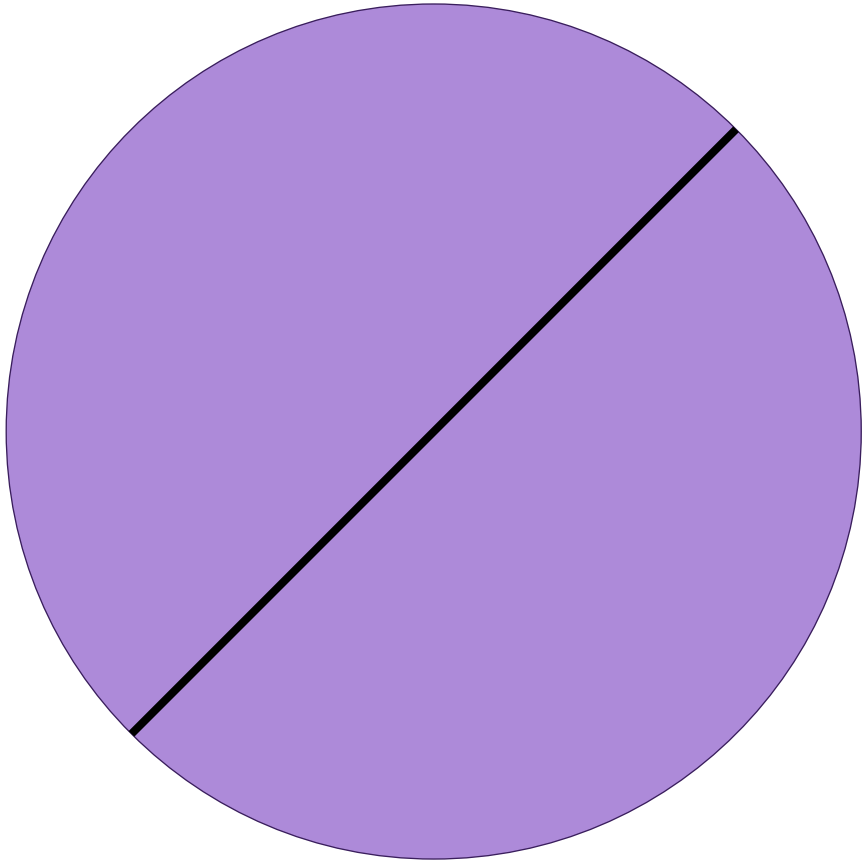
Computable Numbers

Is $\frac{1}{3}$ *computable?* 0.333.....

Is τ computable?

3.14159 2653 . . . . . .

$$\pi = \frac{C}{d}$$



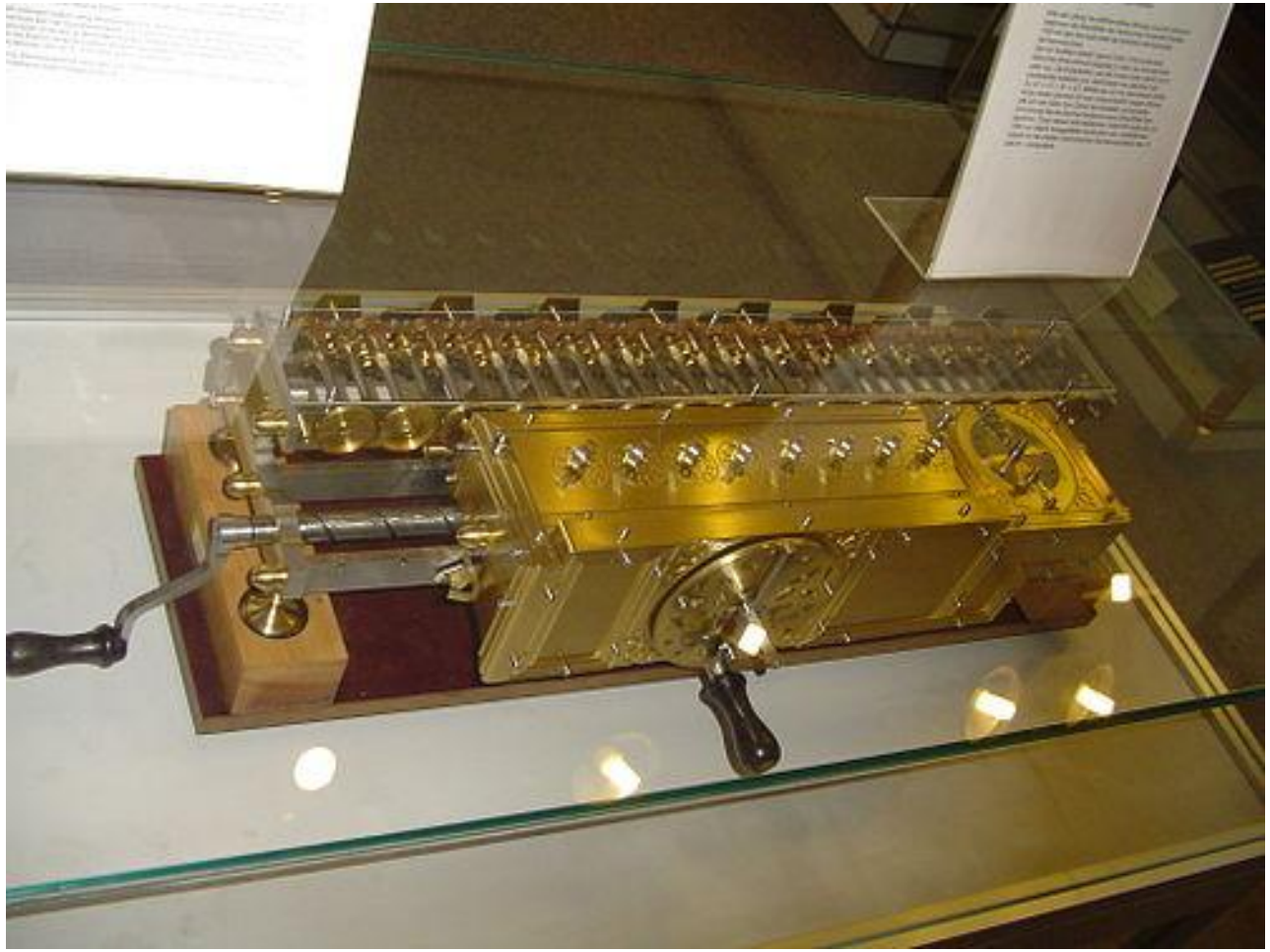
$i = 200M$

$\pi[i]$

$$\pi = \frac{C}{d}$$

$$\pi = 4 - \frac{4}{3} + \frac{4}{5} - \frac{4}{7} + \frac{4}{9} - \frac{4}{11} + \cdots$$

1673



All four basic arithmetic, + - * /
https://en.wikipedia.org/wiki/Stepped_reckoner

Logistics

- Midterm2 next Tuesday, Mar 24
- Free food/drink next week? (at Greenberry's?)

HW 6 due next Monday, Mar 23.

Quiz 8 due next Monday, Mar 23.

Plan

Computability

Textbook [TCS] Section 9

https://introtcs.org/public/lec_08_uncomputability.html
Universality and uncomputability